Lesson notes · for you

# Attention and the transformer

University, 4

Materials
A phone with Darsi; Paper for one hand-worked step; The slides from the kit

## Learning goals

- Score, then mix
- Add the original back

## Activity: In the lab

1. Read the tune
2. Read the next four notes

## Discussion

- What was not counted?

## Check yourself

- Attention replaces which hope?

**Try it this week**
Write one sentence: what the Eight notes lab counted, and what a transformer trained on many tunes would use that the lab cannot (every earlier note, not only the last).

Worksheet

# Attention and the transformer

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Name | Class | Date

**1** **Attention gives three items the weights 0.7, 0.2 and 0.1. Their values are 10, 20 and 30. What does it hand on?**

◯ 20 ◯ 14 ◯ 10

**2** **The Eight notes lab composed a piece of music.**

◯ True ◯ False

**3** **Match the piece.**

| | | |
|---|---|---|
| Query | • • • • | What an item hands over, in proportion to its weight |
| Key | • • • • | What this item is looking for |
| Value | • • • • | Adding the original back |
| Residual | • • • • | What each item can be matched on |

**4** **Order an attention step.**

- ☐ Mix the vectors with those weights
- ☐ Score every item
- ☐ Turn scores into weights that add to 1

✎ **What did the eight notes fail to be?**

Check your answers

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**1** **14**

A weighted sum: 0.7 × 10 + 0.2 × 20 + 0.1 × 30 = 7 + 4 + 3 = 14.

**2** **False**

It continued the counts from one short tune, looking only at the last note.

**3** **Query — What this item is looking for**
**Key — What each item can be matched on**
**Value — What an item hands over, in proportion to its weight**
**Residual — Adding the original back**

The query and the keys make the weights, and the values are what gets mixed.

**4** **Top to bottom: 3, 1, 2**

The mix is last. The weights have to exist first.